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Paleomagnetism of Pleistocene Usami Volcano, Izu Penninsula, Japan—Intensity of the Earth's Magnetic Field in Geological Time II*

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Abstract

The natural remanent magnetization of Pleistocene andesite and scoria samples collected from Usami Volcano Izu Penninsula was examined by paleomagnetic methods and found to be stable. The directions of the stable magnetization of these rocks show significant departure from that of the present geomagnetic field in Japan A hypothesis based on geological field evidence that the strata in the sampled area have been tilted sometime after the formation of the rocks, can account for this fairly well. Using these samples with stable remanences, the intensity of the Pleistocene geomagnetic field (normal and reversed) in Japan has been determined by means of Thellier's metood. Many samples show some irreversible changes in magnetic properties when they are heated above certain critical temperatures. The intensity estimates were obtained from the linear relation of J_n-J_t in the temperature range where irreversible changes did not occur. The intensity of the Pleistocene geomagnetic field determined in the present study is very similar to the present field intensity in Japan (0.46 œ), the mean values being 0.61±0.61 œ for reversed samples and 0.55±0.16 œ for normal ones.

1. Introduction

Kono and Nagata (1966, 1968) obtained 17 estimates of the geomagnetic field intensity in late Pliocene by applying Thellier's method (*Thellier and Thllier*, 1959) to basaltic rocks from two distinct lava flow successions in the southwestern U.S.A. The results of their study can be summarized as follows. (1) The range of variation of the intensity is 0.35–0.82 œ. The intensity values for late Pliocene do not differ from that of the present geomagnetic field at the sampling sites (0.52–0.53 œ) by more than a factor of two. (2) There seems to be little systematic difference between the intensities obtained from normal and reversed samples, although normal field data are few (three intensity estimates only). They also emphasized that intensity data from reliable lava flow sequences are very important for the study of the time-dependent behavior of the geomagnetic field.

The geology and petrology of Pleistocene volcanic rocks in the Hakone-Izu area, Japan, have been extensively studied by *Kuno* (1950, 1951), and the volcanic history of this area is fairly well known. A paleomagnetic study of these rocks (*Nagata et al.*, 1957) indicates

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that the natural remanent magnetization (NRM) of these rocks is stable in most samples and the polarity of NRM is normal except in some of the oldest rocks (Usami Volcano andesites). In the Usami Volcanic Field, the upper and the lower lava flows are also normally magnetized, whereas the middle flows have reversed NRM. *Nagata et al.* (1957) carried out detailed investigations of the NRM and the magnetic properties of these rocks and concluded that this reversed NRM was due to a field reversal, possibly the last reversed field.

Recently, with the construction of a new highway in the Izu Penninsula, fresh outcrops of andesite lavas and volcanic deposits have been exposed in the Usami Volcanic Field. In a preliminary reconnaissance survey with a portable fluxgate magnetometer, it was observed that these newly exposed lavas form a clear succession and have reversed (upper layers) and normal NRM (lower layers). Accordingly, an extensive sampling was carried out and samples were collected from these lave flows and a layer of scoria deposit.

In order to obtain reliable intensity estimates, the stability of NRM and the properties of the ferromagnetic minerals in these rocks were first investigated by the usual paleomagnetic methods. The field intensity in Pleistocene in Japan was then determined by applying Thellier's method to samples whose NRM was stable.

2. Geological Description of the samples

A series of activities of strato-volcanoes took place in Pleistocene and lava flows and volcanic ejecta from these volcanoes cover wide area in Hakone-Izu Volcanic Region (*Knuo*, 1950, 1951). The oldest activity in this area is that of Usami Volcano and after that the center of volcanic activity migrated successively northward as time elapsed; i.e., Taga Volcano, Yugawara Volcano and lastly Hakone Volcano (Fig. 1).

Sampling sites are shown in Fig. 2. The location of samples collected in the previous study (*Nagata el. al.*, 1957) is also indicated in this figure. Lava flows sampled in the present study are considered to be stratigraphically near or a little lower than the lavas sampled by *Nagata et al.* (1957). This geological observation (H. Kuno, personal communication) is consistent with the paleomagnetic direction data; the upper and the lower sections were normal and the middle section was reversed in the previous study, whereas normal flows appear below reversed flows in the present study.

A schematic representation of the sampled sequence is given in Fig. 3. The section is composed of andesite lava flows and pyroclastic or air fall deposits. The thickness of a lava is about three to five meters. These layers are inclined about 20 degrees northeastward from the horizontal plane with the strike axis in NW-SE direction. This inclination is rather peculiar as the probable eruptive vent of Usami Volcano is situated to the northeast of the sampling site (Fig. 2). It is geologically very probable that a tilting of the strata occurred after the emplacement of these volcanics (H. Kuno, personal communication). The detail of this hypothesis will be discussed in the next section.

As the rocks have been exposed very recently, the outcrops are fairly fresh and the order of succession of layers is very clear (Fig. 3). A tuffaceous layer andwiched between the lavas UV10 and UV11 is slightly weathered, indicating a time-gap of the order of 10^2–10^3

years between these two lavas (H. Kuno, personal communication). Other layers do not show any weathered interfaces.

Five to nine core samples (diameter 2.45 cm, length 10–15 cm) were drilled from each lava or scoria layer by means of a portable engine drill (*Doell and Cox*, 1965). A Brunton compass was used to orient the samples. Two specimens of 2.3 cm long were cut from each sample core and one was used for NRM and AC demagnetization measurements. The other was kept for the intensity determination by Thellier's method. Remanences were measured with an astatic magnetometer.

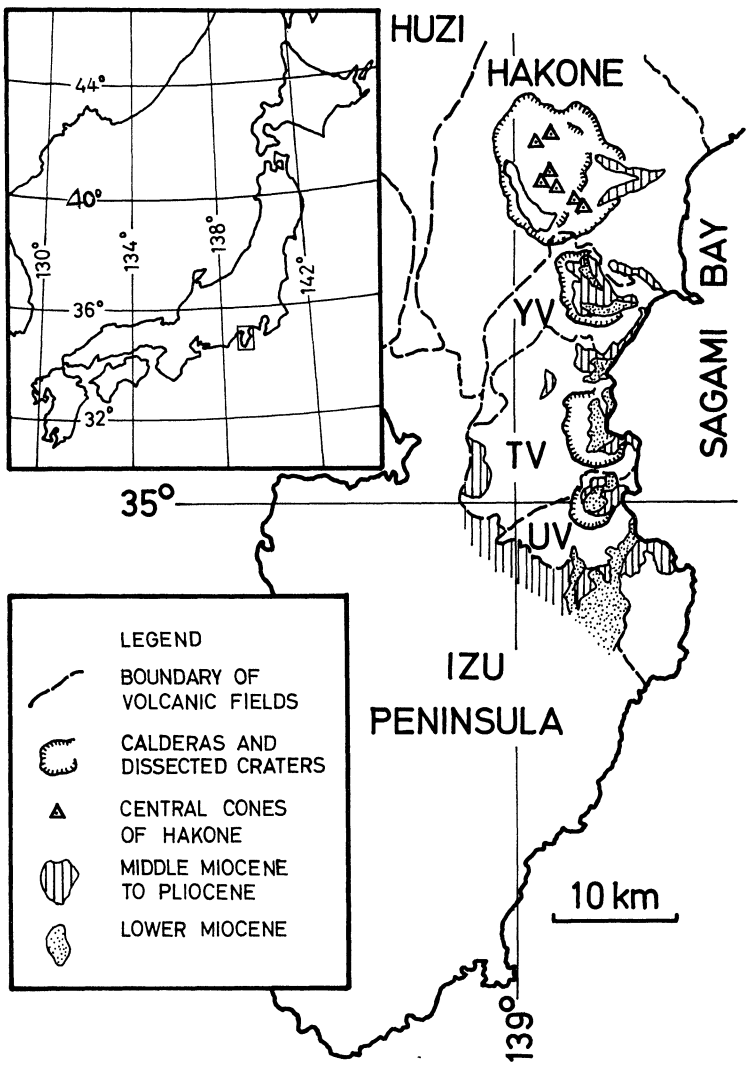


Fig. 1 Geological map of Hakone-Izu Volcanic Region, Japan, redrawn after *Kuno* (1950). UV, Usami Volcano; TV, Taga Volcano; YV, Yugawara Volcano.

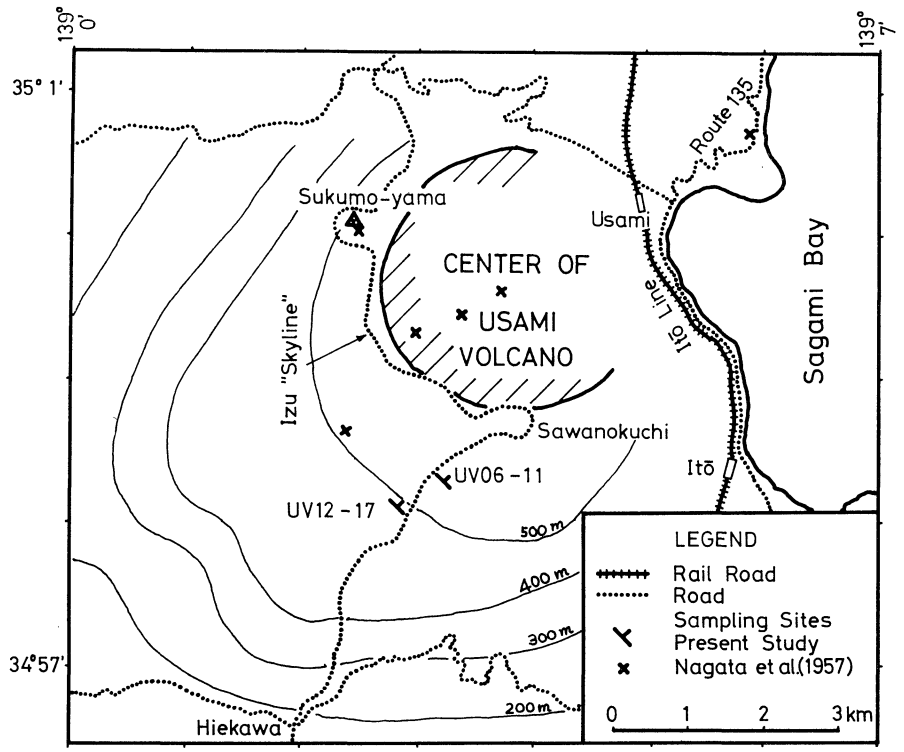


Fig. 2 Sampling sites in Usami Volcania Field. Contour lines show the original form of Usami Volcano before erosion and tilting occurred (H. Kuno, personal communication).

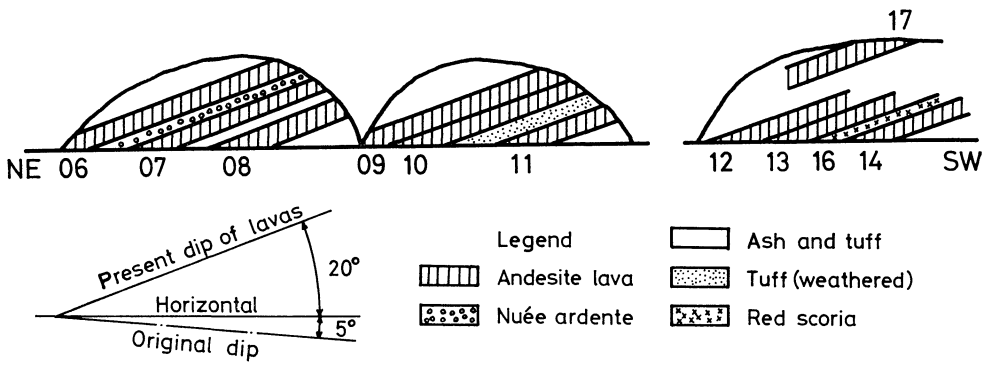


Fig. 3 Schematic representation of the sampled sequence and the relation between the original dip and the present dip of these layers.

3. The Direction and Stability of NRM

The NRM of lavas UV06–UV10 is reversed (up dip), whereas NRM of lavas UV11–UV14, UV17 and scoria layer UV16 is normal (down dip). NRM direction and intensity data are summarized in Table 1.

Table 1 Summary of Paleomagnetic Direction Data of Pleistocene Usami Volcano

Layer Number	Original Magnetization (NRM)						Stable Component after AC Demagnetization (120 œ)				
	N	I	D	k	α_{95}	J_N	N	I	D	k	α_{95}
UV06	5	−52.9	79.0	86	8.3	4.48	5	−58.8	89.2	134	6.6
UV07	7	−48.1	107.8	100	6.1	9.04	6	−53.7	117.7	223	4.5
UV08	5	−42.6	99.5	83	8.4	6.23	5	−49.7	109.3	56	10.3
UV09	6	−13.5	93.8	3.9	39.0	3.70	5	−39.0	114.1	37	11.2
UV10	8	−9.7	91.9	4.5	29.3	4.35	8	−38.7	113.4	57	7.4
UV11	5	63.0	304.8	109	7.4	13.4	5	63.4	291.4	122	6.9
UV17	5	56.5	339.4	42	11.9	5.62	5	56.7	320.9	55	10.4
UV12	9	56.8	314.1	17	12.7	9.56	8	53.1	313.8	26	11.1
UV13	9	61.2	327.3	48	6.7	9.35	9	61.1	326.1	58	6.1
UV16	7	49.5	232.8	86	5.7	5.62	7	48.4	332.7	65	6.6
UV14	9	58.9	324.5	179	3.8	14.8	9	59.4	322.4	99	5.2

N: Number of samples.
I: Inclination (dewnward positive) in degrees.
D: Declination eastward from north, in degrees.
k: Fisher's (1953) precision parameter.
α_{95}: Semiangle of cone of 95% confidence for mean direction in degrees.
J_N: Average intensity of original NRM in 10^{-4} emu/gr.

After the measurement of NRM, all the specimens were left in the geomagnetic field in the laboratory with random orientations. After about fifty days, the magnetization of each speciman was again measured (storage test). The directional changes between the measurements ware less than 3 degrees for most specimens (56 out of 75 specimens). Among the rest, the changes were less than 6 degrees for 12 specimens, and only 7 specimens showed changes of 7–15 degrees. The intensity of the remanent magnetization of a specimen does not change singnificantly between measurements. Thus the NRM of these rocks is stable to strorage test. About two specimens from each layer were subjected to a stepwise AC demagnetization, with peak AC fields of 30, 60, 120, 240 and 480 œ. Of the 22 specimens, UV1006 and UV1203 were unstable to AC demagnetization and the stable component could not be isolated even after demagnetization by 480 œ (UV1006 means the specimen cut from the sixth sample core of the layer UV10). The rest showed comparatively small changes in the direction of magnetization. Examples of decay modes of NRM with increasing AC field are illustrated in Fig. 4. All the rest of the specimens (53) were demagnetized with a peak AC field of 120 œ, which was sufficient to eliminate soft components in most specimens. Average directions of the stable component (magnetization measured after AC demagnetization by 120 œ) are summarized in Table 1. Although dispersion of direction is much reduced in the cases of UV09 and UV10, average directions of magnetization and statistical dispersion parameters (k, α_{95}) for the other layers are almost unchanged by AC demagnetization (Table 1).

The NRM in these rocks, therefore, is stable and can be taken as representative of the ambient geomagnetic field when these rocks were formed. According to the experimental

results (especially those of the thermal demagnetization and analyses of ferromagnetic minerals), it seems very unlikely that self-reversal has occurred in any of the samples used in this study.

The avarage directions of magnetization, however, are significantly different from either that of the present earth's field or from that of an axial dipole field. The hypothesis of landmass movement explained in Section 2 can account, to a considerable degree, for this

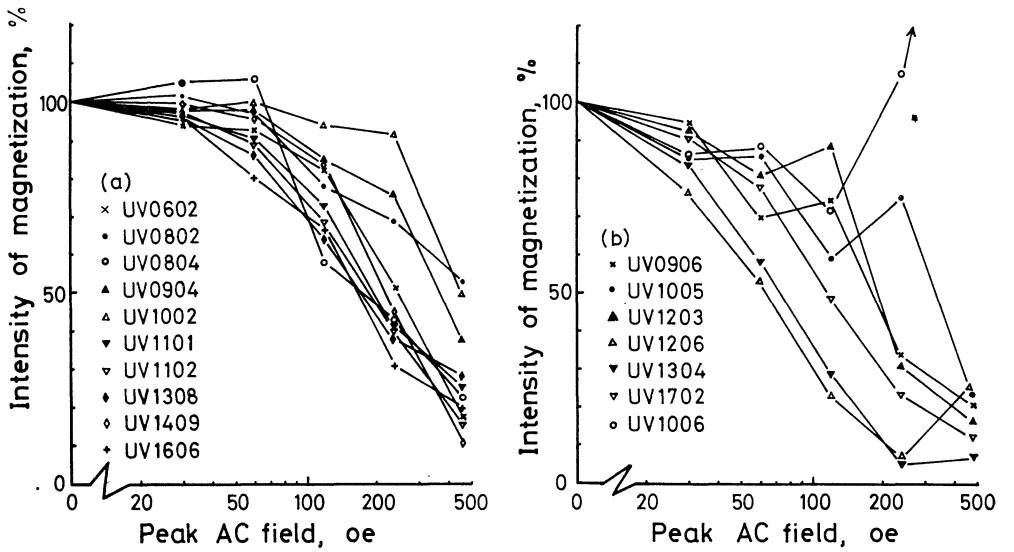


Fig. 4 Examples of the change of intensity of remanent magnetization by AC demagnetization. (a) Stable samples whose direction of magnetization did not change more than five degrees. (b) Samples with direction changes of more than thirty degrees.

Table 2 Directions of Stable Magnetization and Virtual Geomagnetic Poles after Bedding Correction

Layer Number	Magnetization after Bedding Correction			Virtual Geomagnetic Pole Position			
	I	D	α_{95}	Latitude	Longitude	dp	dm
UV06	−69.3	139.3	6.6	−56.2	274.3	9.6	11.3
UV07	−53.7	152.5	4.5	−67.4	235.4	4.4	6.3
UV08	−54.3	141.2	10.3	−58.4	240.2	10.2	14.5
UV09	−43.6	136.1	11.2	−51.2	227.3	8.7	14.0
UV10	−43.5	134.8	7.4	−50.1	227.8	5.7	9.2
UV11	62.6	342.7	6.9	74.0	88.5	8.4	10.8
UV17	47.1	351.4	10.4	80.1	9.1	8.7	13.5
UV12	46.7	343.2	11.1	74.1	27.7	9.2	14.3
UV13	49.0	358.6	6.1	84.8	332.6	5.3	8.1
UV16	36.7	353.1	6.6	74.3	343.7	4.5	7.7
UV14	49.2	354.6	5.2	83.3	3.6	4.6	6.9

I, D, α_{95}: Same as in Table 1.
Pole position: Expressed in northern latitude and eastward longitude, in degrees.
dp, dm: Semiaxes of oval of 95% confidence level for virtual pole, in degrees.

discrepancy. The strata were probably inclined about five degrees south-westward at the time of the emplacement of lavas (see original shape of Usami Volcano in Fig. 2) and afterwards tilted to the present inclination (Fig. 3). As the NRM of these layers is acquired as thermoremanent magnetization (TRM) and is stable to AC demagnetization, we may assume that the NRM has not changed its direction in this tilting or in subesquent geologic time. The direction of magnetization after this tilt correction becomes nearer to that of the axial field so that the virtual poles distribute near the rotational axis (Fig. 5) in accordance with the previous paleomagnetic study of Usami Volcano (*Nagata et al.*, 1957).

The scoria layer UV16 is about one meter thick and underlies the lava UV13 (Fig. 3). This layer if reddish in color in the upper portion (about forty centimeters) and brownish in the lower part. As the lower part was very fragile, samples were collected only form the upper reddish part. Intensity and stability of NRM of this scoria layer were very similar to those of andesite lavas (see Table 1 and Fig. 4). The origin of this stable NRM will be discussed in Section 6.

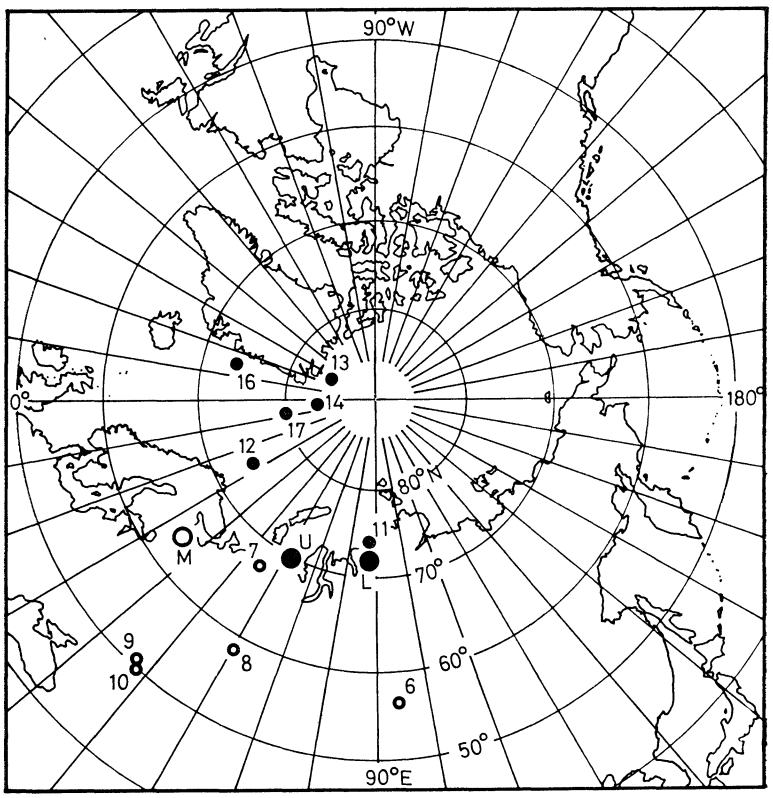


Fig. 5 Distribution of virtual geomagnetic poles determined with the bedding correction from Usami Volcano andesites and scoria. Larger circles show the positions of poles for the upper(U), the middle(M) and the lower(L) strata of Usami andesites determined by *Nagata et al.* (1957). Solid symbols for north-seeking poles, open symbols for south-seeking poles.

4. Analyses of Ferromagnetic Minerals

Thermomognetic analyses were carried out using crushed fragments (1–2 mm diameter) of a smaple from each layer. Variation of saturation magnetization (J_s) with temperature (T) of a sample was measured in a magnetic field of about 2500 α in vacuum of 10^{-3} mmHg by meams of a magnetic balance. Although the highest Curie Points (T_c) of all samples lie in the range of 500–570°C, lower Curie points were observed in several samples (Table 3).

Cooling branches of J_s-T curves of many samples do not coincide with the heating branches. These irreversible changes are probably due to same chemical changes of ferromagnetic mierals, e.g., oxidation, reduction, decomposition, etc., caused by heat-treatment (*Ozima and Larson*, 1967). This is a serious problem in the determination of field intensity by Theller's method, since this method is based upon the assumption that the acquisition process of TRM can be reproduced in laboratory. It is shown in stepwise-heating J_s-T curves (Fig. 6) that the magnetization of an irreversible sample changes reversibly when the temperature of heat treatment does not exceed a certain critical temperature. Above this temperature, some chemical change occurs in the ferromagnetic minerals and the J_s-T curve becomes irreversible (Fig. 6). The critical temperatures are about 350°C for both UV0902 and UV1101. Other samples showed similar but less pronounced irreversible changes in J_s-T curves with oritical temperatures between 300 and 450°C.

X-ray diffraction analyses were carried out for magnetically separated powder of ferromagnetic minerals. Diffraction peaks corresponding to a cubic crystallographic structure were observed in all samples, but no peaks due to a rhombohedral phase (ilmenite-hematite solid solution) could be detected in any sample. Some of the diffraction charts showed two distinct series of peaks indicating the coexistence of two cubic phases with different lattice constants.

Table 3 Properties of Ferromagnetic Minerals in Rocks from Usami Volcano

Layer	Rock Type	Sample	Curie Point(s), °C	Lattice Parameter(s), A
UV06	Andesite	UV0603	(350*), 505	8.43
UV07	Andesite	UV0703	(350*), 500	8.42 8.44*
		UV0702	540	——
UV08	Andesite	UV0803	505	8.38*, 8.44
UV09	Andesite	UV0903	(240), 510*	8.41*, 8.45
		UV0902	(300), 500*	——
UV10	Andesite	UV1002	565	8.37
UV11	Andesite	UV1102	565	8.38
		UV1101	(350*), 565	——
UV17	Andesite	UV1701	500	8.38
UV12	Andesite	UV1203	520	8.37
UV13	Andesite	UV1308	550	8.39
UV16	Scoria	UV1602	(480*), 560	8.37*, 8.41
UV14	Andesite	UV1401	505	8.43

* Minor components.
Values in brackets are approximate.

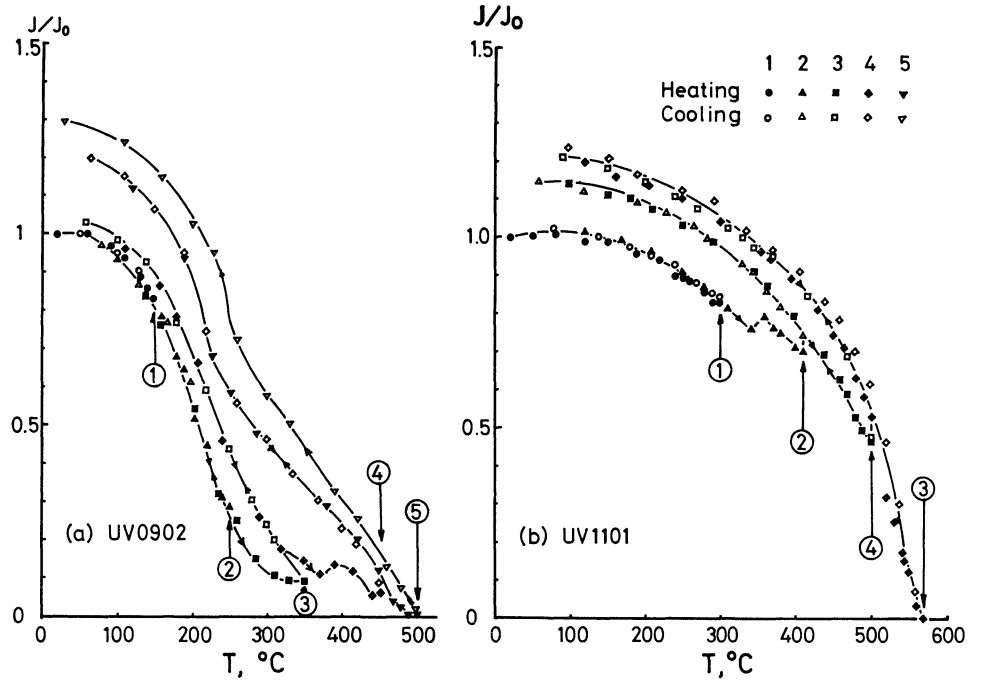


Fig. 6 Examples of changes in J_s, measured in a field of 2500 α in air, during heatings to successively higher temperatures. The highest temperature in each heating step is indicated by a number in a circle. (a) UV0902. (b) UV1101.

Results of thermomagnetic and X-ray diffraction analyses are summarized in Table 3. As the main ferromagnetic component in these samples has cubic structure, a lattice parameter of about 8.4A, and T_c in the range 500–570°C, it is likely a titanomagnetite.

5. Field Intensity Determination by Thellier's Method

From each layer, one sample having an NRM which is stable to AC demagnetization and which also is near to the mean direction of magnetization of the layer was selected for intensity determination. The second specimen (unused in the foregoing experiments) from the selected sample core was used in the Thellier experiments. For the lavas UV07 and UV13, three samples each were selected for intensity specimens to investigate the internal consistency of intensity data. Heating experiments were carried out in air at the Kakioka Geophysical Research Laboratory, University of Tokyo (F_0=0.461 α). Experimental procedures closely follow those described by *Kono and Nagata* (1968).

Some of the results are shown in Fig. 7. In these figures, J_n stands for the NRM component which remains after thermal demagnetization from room temperature to some temperature T, and J_t stands for the partial TRM component produced in the laboratory field (F_0) in the same temperature range. The negative gradient of the J_n-J_t linear realation will be equal to the intensity ratio F/F_0, where F denotes the intensity of the ancient magnetic field.

In all the present samples, the J_n-J_t relation is fairly close to linear over the temperature

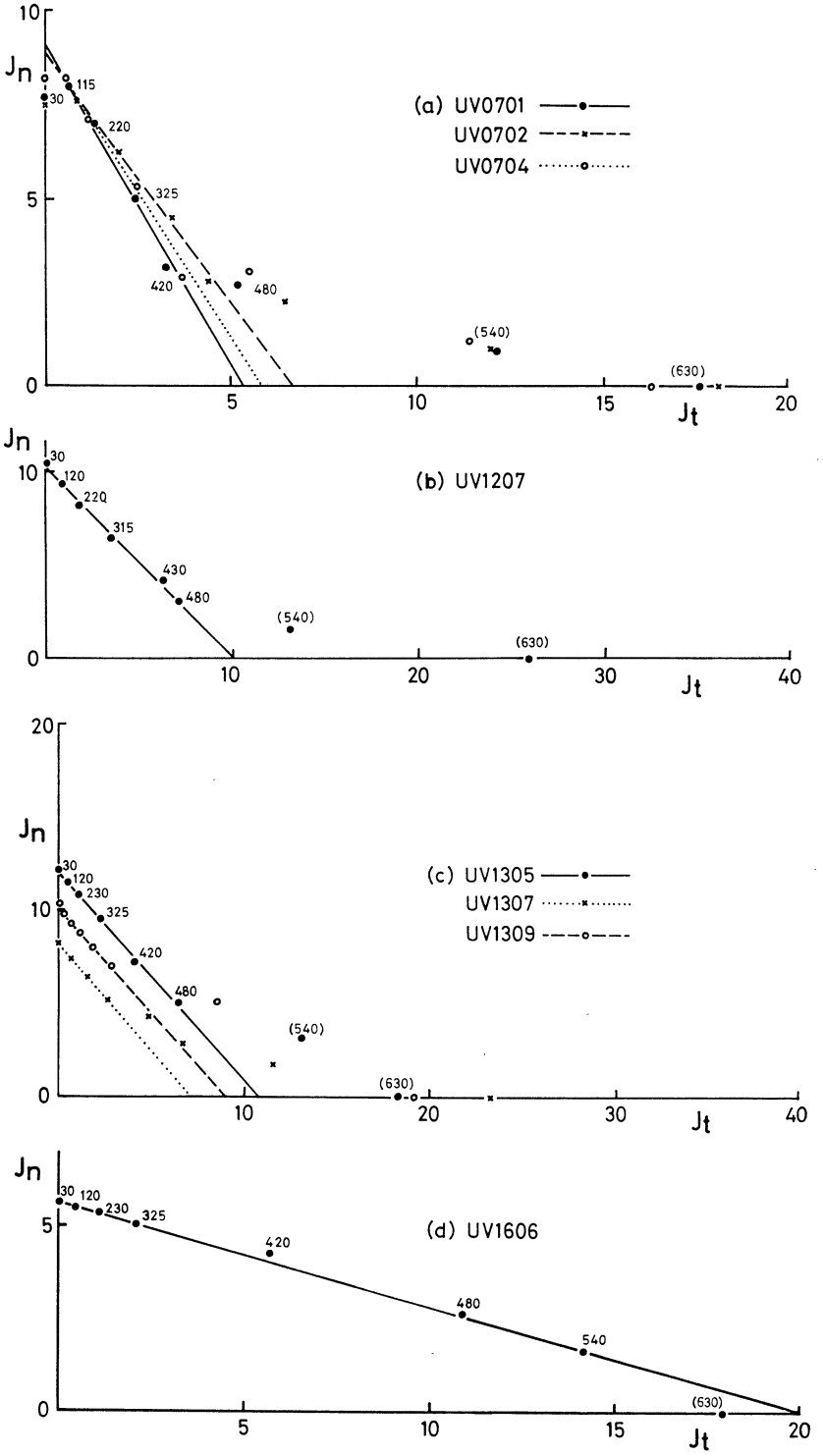


Fig. 7 Results of the Thellier experiments shown as J_n-J_t diagrams. The axes are labelled in units of 10^{-4} emu/gr. and the numbers associated with the points indicate the temperatures in °C of the heating steps. Points with temperatures in brackets have no real NRM component. Values of "J_n" for these points correspond to the amount of increase of TRM component between the two heatings to the same temperature (see text).

range 100°C to 420 or 480°C. The departure of the low-temperature points from this line is apparently due to viscous remanent magnetization (VRM) superimposed on the stable component. The direction of VRM components is parallel to the present geomagnetic field; room temperature points for reversed samples (UV06 through UV10) fall below the lines, while the points for normal samples (UV11 through UV17) lie above the lines. By thermal demagnetization to about 100°C, the VRM component is removed and the direction of the NRM component becomes coincident with that of the stable component, as determined previously by AC demagnetization.

Departure from the J_n-J_t linear relation at higher temperatures (in most cases, higher than 480°C) is considered to be due to irreversible changes of magnetic properties of the sample in the heating process. The NRM component was completely destroyed at the temperatures shown in brackets in Fig. 7. The "J_n" component indicated in Fig. 7 has its direction parallel to the laboratory field and, moreover, 540°C is higher than the Curie point of most samples (Table 3). The "J_n" at these temperatures probably corresponds to the increase of TRM component between the two heatings to the same temperature. Further heating to 630°C produces much greater TRM in many samples, suggesting that the

Table 4 Summary of Intensity Data from Pleistocene Usami Volcano, Japan

Layer	Sample	NRM/TRM	F/F_0	F, œ	VDM$\times 10^{-25}$, gauss cm^3
UV06	UV0605	0.32	0.82	0.37	5.5
UV07	UV0701	0.44	1.73		
	UV0702	0.42	1.41		
	UV0704	0.50	1.56		
	averaged		1.56	0.73	13.5
UV08	UV0802	0.53	1.44	0.67	12.3
UV09	UV0902	0.38	1.00	0.46	9.5
UV10	UV1002	0.52	1.80	0.84	17.4
UV11	UV1101	0.64	1.10	0.515	8.5
UV17	UV1701	0.48	1.80	0.84	16.8
UV12	UV1207	0.41	1.02	0.465	9.3
UV13	UV1305	0.66	1.17		
	UV1307	0.35	1.17		
	UV1309	0.54	1.17		
	averaged		1.17	0.545	10.7
UV16	UV1606	0.32	0.29	0.135	3.0
UV14	UV1409	0.78	0.79	0.37	7.2
Mean of 5 reversed samples			1.32±0.40	0.61±0.19	11.6±4.5
Mean of 5 normal samples (except UV16)			1.18±0.38	0.55±0.16	10.5±3.7
Mean of all data (except UV16)			1.25±0.38	0.58±0.18	11.1±3.9

NRT/TRM: Ratio of intensity of NRM to that of total TRM produced by cooling from 630°C to room temperature in a field of $F_0=0.461$ œ.

VDM: Virtual dipole moment defined by $\mathrm{VDM}=\frac{Fr^3}{2}(1+3\cos^2 I)^{1/2}$ where I and F are inclination and intensity of the ancient geomagnetic field and r is the earth's radius (Smith, 1967).

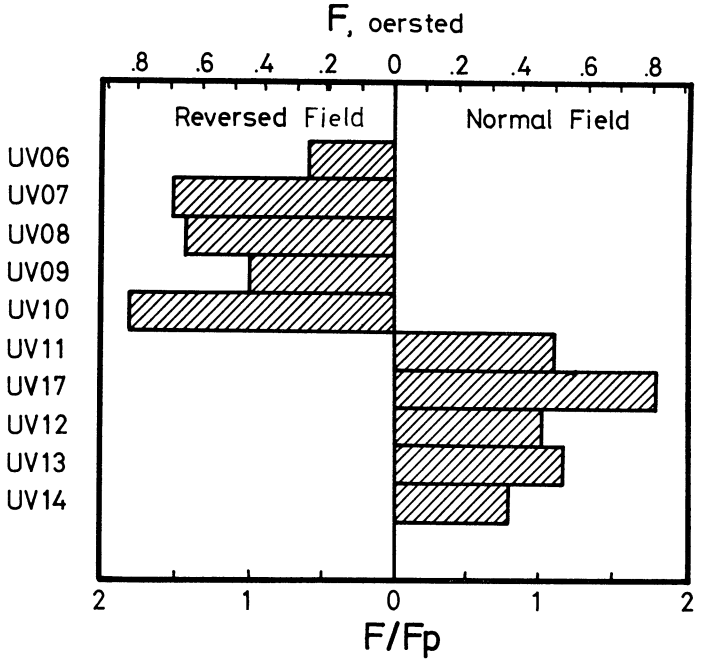


Fig. 8 Change of the field intensity in Pleistocene in Japan determined from Usami Volcano andesites.

changes of magnetic properties are accelerated at elevated temperatures.

It is interesting that the J_n-J_t linear relation seems to hold in many samples up to a temperature about 100°C higher than the critical temperature observed in the J_s-T curves. It seems probable that the chemical conditions such as oxygen fugacity are different in heatings of a massive sample or crushed powders. Detailed investigation of these effects are panned.

The intensity values are determined from the J_n-J_t diagrams neglecting the points affected by VRM at low temperatures or change of magnetic properties at high temperatures. Results of the Thellier experiments are summarized in Table 4 and Fig. 8. The intensity estimates obtained in the present study, except that for the scoria layer UV16, do not differ by more than a factor of two from the present field intensity in Japan (9.46 œ) regardless of the polarity of the geomagnetic field when the rocks were formed. This is also consistent with the conclusions of the intensity study of the late Pliocene field in the southwestern U.S.A. (*Kono and Nagata*, 1966, 1968).

6. Origin of NRM in the Scoria Layer UV16

The reddish part of the scoria layer UV16 has an NRM which is very similar to the NRM of the lavas, both in intensity and in stability of AC demagnetization (Table 1 and Fig. 4). This stable NRM may be acquired by one of the following three mechanisms; (1) the scoria was still hot (higher than T_c of this layer, 560°C) at the time of deposition and acquired TRM during cooling *in situ*, (2) the scoria was heated to a temperature higher

than 560°C by the overlying lava flow UV13 and acquired TRM, or (3) the scoria acquired chemical remanent magnetization (CRM) in some chemical change of ferromagnetic minerals which took place when the lava UV13 heated this layer to a temperature lower than T_c.

According to H. Kuno (personal communication), it is very unlikely from the geological point of view that the temperature of the scoria was still higher than 560°C at the time of deposition at a place very distant from the probable vent of Usami Volcano (about 3 km, see Fig. 2). The first mechanism may therefore be discarded. Also, the directions of stable remanence and the field intensities determined by Thellier's method for UV13 and UV16 differ considerably, making the second mechanism unlikely.

The Thellier's experiment for the sample UV1606 shows that there is a fairly good linear relation between J_n and J_t (Fig. 7d), but the intensity estimete from this sample (0.135 œ) is only about a quarter of the value obtained from UV13 samples (0.545 œ). Kobayashi (1959) showed that CRM generated at 300°C or at 270°C by reduction of α-Fe_2O_3 (hematite) to Fe_3O_4 (magnetite) shaws almost identical behavior to AC and thermal demagnetization as TRM produced in the same sample. The *intensity* of CRM, on the other hand, is about one tenth of total TRM generated in the same magnetic field. It is possible that the NRM of the UV16 layer is a CRM, but it is difficult to be certain of this as the acquistion mechanism of CRM is not fully understood and as, moreover, it cannot be reproduced in the laboratory. Hence UV16 is excluded in the discussion of field intensity. For intensity studies, it is safer to avoid samples which might have acquired CRM.

7. Summary

The stability of NRM of Usami Volcano andesite and scoria has been confirmed by storage test, AC demagnetization and the applicability of Thellier's method to these rocks. The reversed NRM in this sequence is concluded to be due to a field reversal, since no indication of self-reversal was observed during the present study. According to geological and paleomagnetic evidence, it is highly probable that some tilting of strata occurred in this area after the volcanic rocks were emplaced and acquired stable NRM. The direction of magnetization after tilt correction is nearer to that of the present geomagnetic field than the direction of NRM *in situ*.

Field intensity estimates obtained from these samples by means of Thellier's method are similar to that of the present geomagnetic field in Japan, for both normal and reversed samples. Irreversible changes of magnetic properties occurred in most samples when the heating temperature exceeded some critical temperature in the range 350–450°C. It is recommended, therefore, that field intensity should be determined from the linear J_n-J_t relation in the temperature range where the magnetization of a sample is not affected by VRM components or by irreversible changes in the magnetic minerals.

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